Concept Review

Servo Motors

Why Servo Motors?

For many mechatronic applications, it is necessary to be able to control the position of an actuator with precision. Unlike open-loop controllers which have no feedback on the state of the motor, closed-loop leverages a position sensor attached to the motor to provide information about the required control signal to drive the motor to the desired position. That is the case with a servo motor where a commanded position is achieved through a controller, which drives the motor based on feedback from a position sensor, such as a potentiometer. The key feature that distinguishes servo motors from other types of motors is their built-in feedback mechanism. They tend to be chosen for applications that require accurate and reliable position control.

1. Introduction

A servomechanism is a device that uses a feedback sensor to automatically adjust the behavior of a system using feedback control. A DC servo motor (often colloquially referred to simply as a servo) is a rotary actuator that transforms electrical energy into mechanical energy and incorporates a feedback mechanism for position control. It holds the commanded position, even under load, until it is instructed to do otherwise. Because they operate in closed-loop control, they can correct disturbances or mechanical load changes by comparing measured versus desired positions.

2. Basic Construction and Operation

A typical servo consists of four components: a motor (which tends to be a DC motor for hobby servos), a reduction gearbox, a position sensor (typically a potentiometer), and a control circuit. These 4 components can be seen in Figure 1.

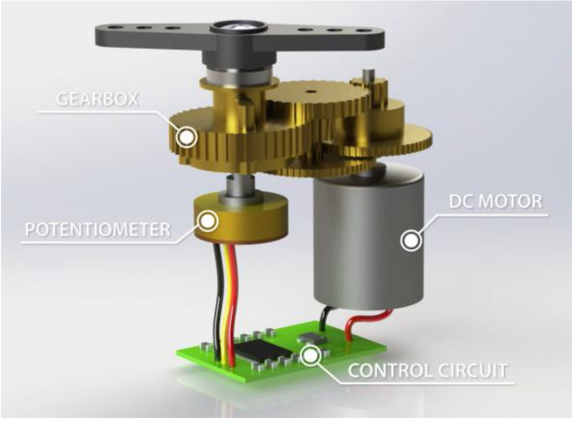

Figure 1. Internal Layout of a Servomotor

The gearbox is used to convert the high speed, low torque motor into a high torque, lower speed output, thus having a high holding torque compared to standalone motors of a similar size.

The potentiometer is used to get information about the current position of the output shaft. This information, plus the control circuit, which receives an input signal of the desired position, allows for closed-loop control.

The control circuit monitors the real time position data from the potentiometer and compares it with the desired position signal; the controller then uses this error between measured and desired position to adjust the input voltage to the motor to move it until the error is minimized and the servo is at the desired position. This allows the servo to keep its position even as load changes dynamically, making them ideal for precise applications. Some systems also use PID control algorithms to optimize accuracy, speed, and stability.

Most servos allow for position control with a limited range, usually around 180˚, however, some have a larger or smaller range, consult the motor's datasheet for more information. However, some servos are made for continuous rotation, like a DC motor, where the control signal is for speed control instead of position control like described above.

a. Wiring

The wires of hobby servos tend to look like Figure 2. Servos have three cables going into the connector: power, ground and signal. The power cable is usually the middle connector, that way, if the servo is connected backwards, no major damage is incurred by the servo. The ground wire is usually either brown or black. The signal wire is usually orange, but can vary between brands to be yellow, white or even blue. Confirm your servo wiring diagram based on your motor's datasheet.

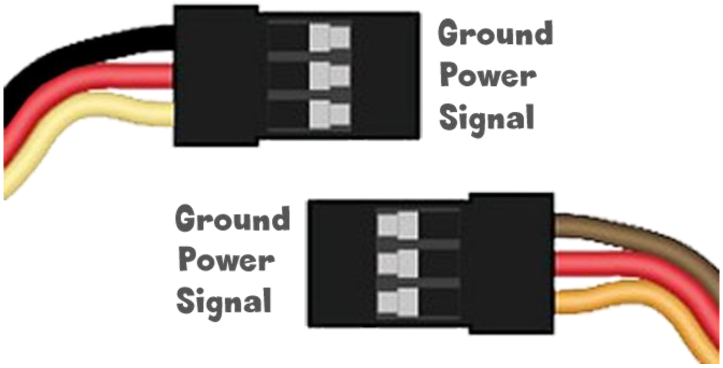

Figure 2. Hobby Servomotors Wiring Connector

3. Driving a Servomotor

The position of a servo motor is generally commanded using a series of pulses using a PWM (pulse width modulation) signal. The servo expects a pulse every 20 milliseconds, which is at a frequency of 50 Hz. The motor's internal circuitry interprets the pulse width and moves the shaft accordingly to reach the desired position.

Whenever the servo is pulsed with a low pulse width, also called Minimum Pulse, the motor will move to the minimum angle. A neutral pulse will result in a rotation of half the range, and a large pulse, also referred to as Maximum Pulse, will result in a rotation to the full range of the servo. The amplitude, or voltage, of this pulse depends on the servo's operating voltage, which tends to be a range which typically includes 5V as part of that range. The duration, or width, of the pulse determines the position of the servo. Hobby type servos tend to only have a range of about 180˚. The low pulse is usually a 1 ms pulse for a position at 0˚, the neutral pulse is a 1.5 ms pulse for a position at 90˚, and the large pulse is a 2 ms pulse for a 180˚ position. However, the minimum and maximum pulse could vary depending on the servo, some servos, like the one provided with the actuators trainer, accept a pulse between 0.5 ms to 2.5 ms as shown in Figure 3.

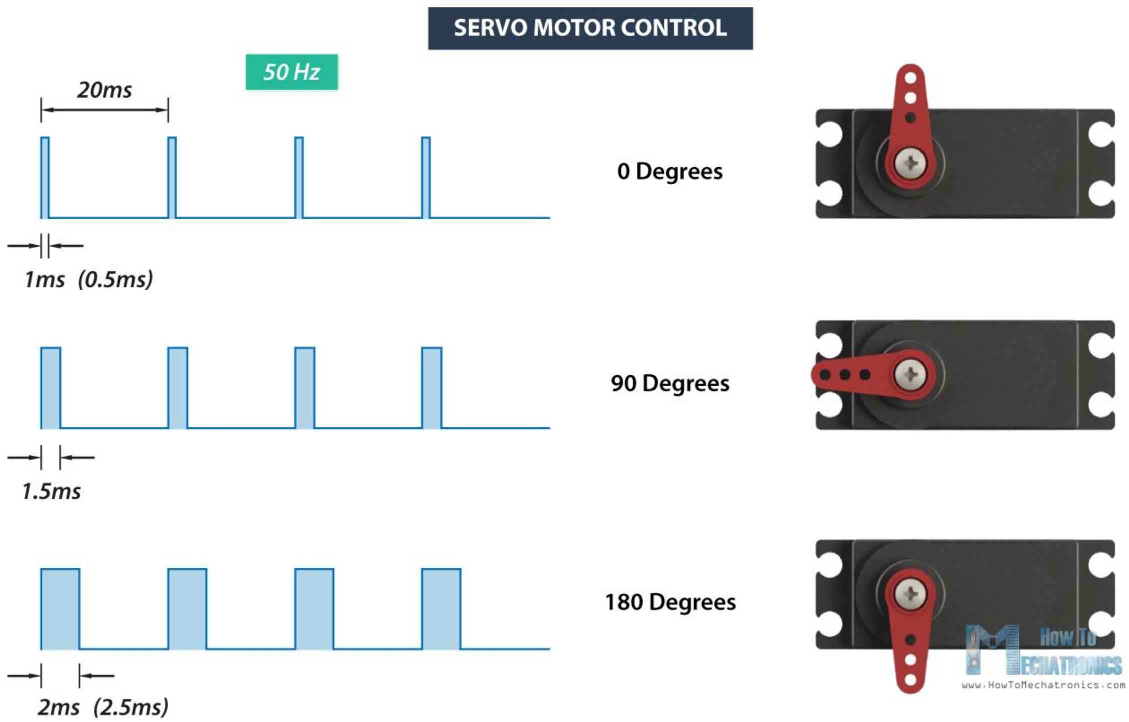

Figure 3. Control Signal for a Servomotor

Note that if you use a 1-2 ms pulse range for a servo that accepts 0.5 to 2.5 ms range, the servo might not be able to get to all its possible positions, but it will not get damaged. The same cannot be said about the opposite, using a 0.5 to 2.5 ms range for a servo that only accepts a 1-2 ms pulse might damage or even break the servo.

4. Datasheet And Considerations

The motor's datasheet summarizes its key electrical and mechanical characteristics and are essential when selecting a motor for a project or application.

a. Understanding a Servomotor Datasheet

When selecting a servomotor, there are certain parts of the datasheet that might be important when looking for motors for a project. This section will mostly focus on the electrical information, however, note that there will also be mechanical information about the motor, usually in the form of a mechanical drawing. This information is important for deciding if a motor will physically fit in a required space, or mount effectively.

When selecting a servomotor, there are several pieces of specification data which will be of particular interest. Figure 4 shows two excerpts from the datasheet for the servomotor provided with the actuators trainer.

4.电气特性 Electrical Specification					
No	Item	Specifcation			
4.1	工作电压 Operat voltage	4.8V	6.0V	7.2V	8.4V
4.2	静态电流 Idle current	＜40mA			
4.3	空载转速 No load speed	0.10sec/60°	0.09sec/60°	0.08sec/60°	0.07sec/60°
4.4	空载电流 Runnig current	210mA	260mA	350mA	450mA
4.5	堵转扭矩 Peak stall torque	12kg.cm	14.5kg.cm	15.5kg.cm	16.5kg.cm
4.6	堵转电流 Stall current	2700mA±10%	3200mA±10%	3900mA±10%	4300mA±10%

5.控制特性 Control Specification		
No	Item	Specifcation
5.1	控制信号 Command signal	Pulse width modification
5.2	放大器类型 Amplifier type	Digital controller
5.3	脉冲宽度范围 Pulse width range	500～2500usec
5.4	中点位置 Neutral position	1500usec
5.5	旋转角度 Running degree	180°±3°(when 500～2500usec)
5.6	死区宽度 Dead band width	3 usec

Figure 4. Servomotor Datasheet Excerpt

This is typical of the sort of data generally available for analog servos. Key information which should be provided for a servo motor includes:

1. Operating Voltage
2. Maximum (stall) torque
3. Maximum (stall) current
4. No Load Speed
5. Deadband
6. Resolution

Given the specification data discussed above, there are several design criteria which will likely be important for deciding whether or not a motor is appropriate for your application.

i. Rated Voltage

This is the voltage at which the motor is designed to run, and at which the motor will have been tested for other specifications. If the motor voltage is significantly greater than that of the power supply driving it, the system will be unable to produce the rated torque. If the supply voltage exceeds the rated voltage, there is a risk of damage to the motor hardware. Most hobby servos operate at 4.8–7.4 V, but industrial versions can handle higher ranges.

ii. Peak Current

This is the current which the motor will draw. For a servo motor, the peak current will be drawn when the servo is exerting peak torque. This will occur when commanding a large step in angle, or when the servo is resisting a large external torque. Current draw varies with load and activity—peak current occurs during acceleration or when resisting force. Some designs benefit from separate power sources for the motor and control electronics to avoid brownouts or signal instability.

iii. Max/Holding Torque

Most applications will have a torque requirement either to actuate a particular mechanical assembly, or to hold position in the face of a specified external load. This is the highest load the servo can resist or apply. Exceeding it causes position errors, but the servo will typically return to its target once the load is removed.

Servo motors are usually rated in kg/cm (kilogram per centimeter), and most hobby servos like the SG90, MG995 etc, are rated between 3kg/cm to 12kg/cm. This rating indicates the weight the motor can lift at a specific distance. For example, a 25kg/cm servo can lift 25kg when the load is 1cm from the motor shaft as shown in Figure 5. The torque at each position is the torque divided by the distance in cm.

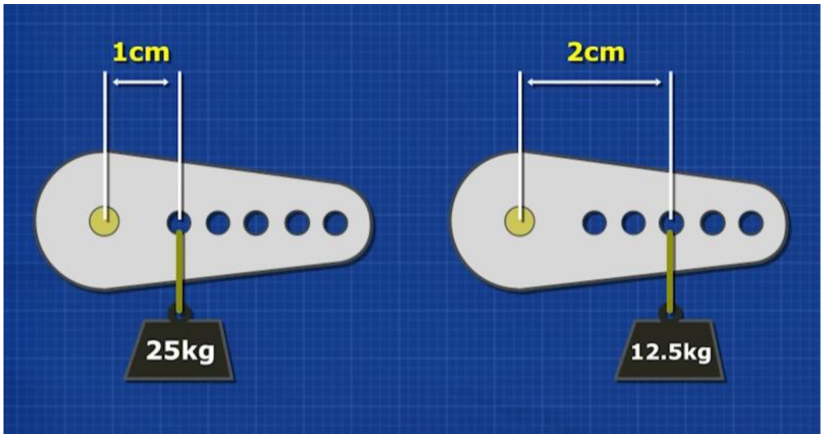

Figure 5. Motor Torque measured in kg/cm

iv. Servo Speed

Servo speed is usually given as the time to move a fixed angle (e.g., 0.1 s per 60˚). This gives a practical measure of responsiveness rather than raw RPM. Datasheets will usually define this along at what voltage this was measured at, at higher voltages, the speed at which the servo moves tends to be faster.

v. Range

The range of a servo motor refers to how far it can rotate from its neutral position. Most hobby servos provide a range of 180˚, although some models support extended motion up to 270˚, or even continuous rotation. The required range depends on the mechanical setup—some applications only need limited angular motion, while others benefit from wider or unrestricted travel.

b. Design Decisions

i. Cost

Cost will often be the principal driver in a design. The final cost of a motor will depend on a large number of variables, including volume of production, and the location of the manufacturer. Hobby servos are low-cost and widely available, while industrial-grade models are more expensive due to higher torque, precision, and durability.

ii. Efficiency

The efficiency of servos is mostly a concern in terms of the amount of heat which is generated as a byproduct of normal operation. Servo motors are extremely efficient under low/no load, they also only use power when correcting position; however, they may heat up when subjected to high variable torques as the control circuitry must constantly drive high currents through the motor coils to resist the applied torque.

iii. Ease of Use

This is harder to quantify based on technical documentation; however, the overriding factor in ease of use is the complexity of the motor control circuitry. However, most servos are simple to use with standard PWM signals and require no external control circuitry. More advanced models may use digital protocols and require configuration.

iv. Lifespan

The lifespan of the motor will sometimes be given in the datasheet either in terms of hours of operation or number of rotations. Servos with brushed DC motors or plastic gears wear over time. Lifespan varies based on use, load, and component quality. Datasheets may list the expected service life.

v. Range of Motion

Typical positional servos rotate up to 180°, with some models offering extended or continuous rotation. While we have focused principally on servo motors with a limited range of motion, there are also free-turning servos available on the market. However, at a given price point and for a given torque and power requirement, if continuous motion in one direction is required, a different type of motor will be a better option.

vi. Position Control Accuracy

This is an area where the servo motor has significant advantages over other motors. Servos provide reliable position control through internal feedback. Gear reduction and PID control ensure good repeatability, even under varying loads without the loss of torque.

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